



# Modeling and analysis of a 3-DOF Lorentz-force-driven planar motion stage for nanopositioning

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## ARTICLE INFO

### Article history:

Received 21 September 2008

Accepted 9 May 2010

### Keywords:

Nanopositioning  
Planar motion stage  
Lorentz motor  
Air-bearing  
Master–slave control

## ABSTRACT

This paper presents an approach to dynamic modeling and analysis of a 3-DOF Lorentz-force-driven planar motion stage for nanopositioning. The planar motion stage consists of a long stroke motor, a short stroke motor and a positioning platen. The long stroke motor is a linear motor, and the short stroke motor is a planar motor propelled by linear Lorentz forces and floated with high-stiffness air-bearings. The Lorentz-force-driven planar motion stage is regarded as two rigid bodies which are modeled by the Newtonian approach independently, then the relations of Lorentz forces with electric currents, magnetic fields, and displacements are obtained through electromagnetic analysis of the short stroke motor using 3-dimension finite element method. The frequency response functions of the planar motion stage in the moving directions are obtained by numerical simulation, which are in good agreement with those obtained experimentally. The positioning accuracy of the planar motion stage is also evaluated numerically and experimentally.

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## 1. Introduction

Nowadays, high speed and high precision positioning stages with multi-degrees of freedom (DOFs) have been increasingly applied in semiconductor manufacturing systems, precision manipulation systems and precision measurement systems, where nanometer, even subnanometer, positioning accuracy and meters per second motion speeds are required [1,2]. Most of the conventional driving systems cannot meet these needs due to their large masses, inevitable mechanical friction, guide imperfection and crawl phenomena, resulting in poor positioning accuracy at the micrometer level. Therefore, development of novel ultra-precision motion stages has attracted research interests all over the world.

Driving devices using magnetic levitation [3–6], piezoelectric (PZT) actuators [7–9], or Lorentz motors with air-bearings [10–12] applied in ultra-precision planar motion stages were reported in the past decades. Among them, magnetic levitation systems are usually unstable due to the dynamic coupling between its 6-DOFs and the strongly nonlinear relationship between electric currents and displacements. PZT actuators are difficult to model analytically due to their strongly nonlinear behavior. In addition, PZT actuators are always utilized in platforms that can only provide velocities up to a few millimeters per second and displacements up to a few microns. In contrast, ultra-precision positioning stages, which are levitated by high-stiffness air-bearings and driven by

Lorentz forces, have many merits over its counterparts. Air-bearings have many advantages including nearly zero friction, high precision, low heat generation, and little contamination [13], while Lorentz motors can generate motions with low-noise, high power density, and large strokes [14]. Moreover, Lorentz motors have good closed-loop controllability compared to magnetic levitation and PZT actuators. Therefore, Lorentz motors with air-bearings are well-recognized as a good choice in advanced manufacturing fields, especially for the tasks requiring high positioning precision with long strokes.

However, there are still many practical challenges to the Lorentz-force-driven planar motion stage (LPMS). The coupled actuating forces along motion directions and non-uniform magnetic fields induce fluctuation of Lorentz forces, which consequently influences positioning accuracy of the LPMS. Therefore, to achieve contented positioning accuracy, exact dynamic model and control strategy are essential to compensate its nonlinearities.

Many works involving dynamic modeling, analysis and experiments for ultra-precision positioning stages have been reported, which can be roughly classified into experimental and analytical approaches. Ziegler and Nichols [15] developed an experimental method which avoids the need for modeling the control path and relies on parameters identified from a step response. In [16], Basilio and Matos proposed an improved tuning method for proportional-integral (PI) and PID. The methodology encompasses the design of PID controllers with under-damped step responses. A novel linear motor-driven aerostatic planar motion table system was developed by Shinno et al. and the full closed-loop control system with

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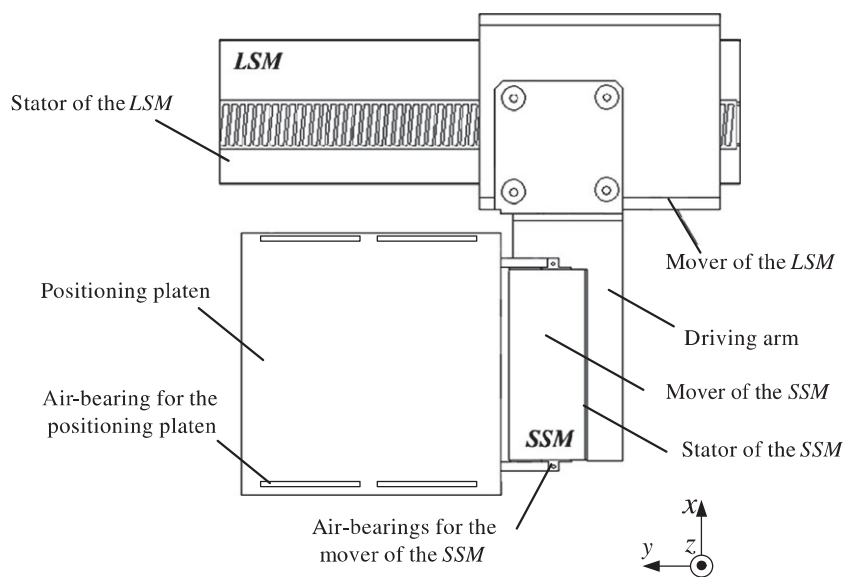


Fig. 1. Basic configuration of the LPMS.

the laser interferometer feedback was used together with *PID* and acceleration feed-forward compensators [17]. However, in these experimental methods, the controllers introduce a high percentage of overshoot and are not capable of fast positioning.

The analytical modeling method can reflect the dynamic characteristics of the system. Once a dynamic model of the stage is precisely obtained, the desired controller can be synthesized analytically. Lee [12] developed a new 3-DOF Z-tilts micropositioning system using electromagnetic actuators and air-bearings where an analytical equivalent method was employed to establish the relations between the air gap, electromagnetic attraction and static thrust forces of air-bearing. In [18], Kwang and Yoon presented a positioning stage using novel contact-free planar actuators based on magnetic forces and Lorentz forces. The equations of motion and the entire voltage equations are derived using a magnetic circuit approach and the modified Lorentz's law. Jin et al. [19] applied a 3-D equivalent circuit method to analyze the leakage flux of planar motor-driven long stroke motor (PMLSM) and the propulsion fluctuation. Lu et al. [20] discussed transient and dynamic performance of an LIM with the coupled-circuit model, in which the propulsion force, motion speed and the sheet currents are evaluated under various transient conditions. Ebihara and Watanobe [21] presented a surface motor that enables bi-directional motion using only one motor and proposed an approximate three-dimensional finite-element analysis method. In [22] Im proposed the volume integral equation method to calculate the magnetic field of air gap and cogging force for synchronous permanent magnet type planar motors. The planar actuator is analyzed by means of the finite element method in order to compute the magnetic flux density distribution and the forces involved, and performances of the platform were improved [23,24]. These works are focused on dynamic modeling and analysis of planar motion stages with short strokes (usually a few millimeters). However, dynamic modeling and analysis of planar motion stages with large stroke (hundreds of millimeters) as well as nano-precision are rarely reported, and some coupling effects are not considered accurately in the modeling.

The motivation of this paper is to conduct modeling, simulation and analysis of a 3-DOFs LPMS, using dynamics, electromagnetics, and experiments. The rest of this paper is organized as follows. Section 2 introduces the mechanical structure, configuration, and working principle of the LPMS. Equations of motion (EOM) of the

LPMS are derived in Section 3. Section 4 presents numerical simulation and analysis on the relationship among the magnetic fields, electric currents, Lorentz forces and displacements for the short stroke motor. In Section 5 FRFs of the LPMS are obtained by numerical simulation in three motion directions. Mechanical FRFs of three motion directions are obtained by experiments to identify our dynamic model in Section 6. Control systems and control strategies are discussed and positioning accuracy of the stage is evaluated numerically and experimentally in Section 7. Conclusions are drawn in the last section.

## 2. Description of the LPMS

### 2.1. Mechanical structure of LPMS

Fig. 1 shows structural configuration of a LPMS which is used as a prototype of LPMSs in our study, where *X*, *Y* and *R<sub>z</sub>* are its three principal motion directions. The LPMS consists of a long stroke motor (LSM), a short stroke motor (SSM) and a positioning platen.

The LSM is a permanent magnet linear motor, consisting of a row of permanent magnets mounted on the stator and a set of coils mounted on the mover. The motion stroke of the LSM is hundreds

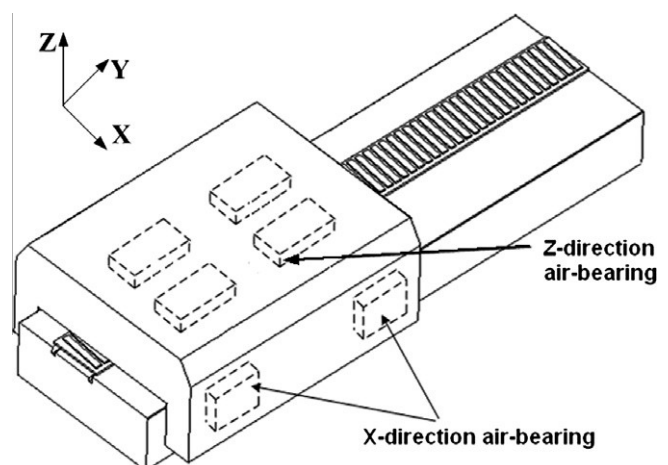


Fig. 2. Schematic of the layout of air-bearings of the LSM.

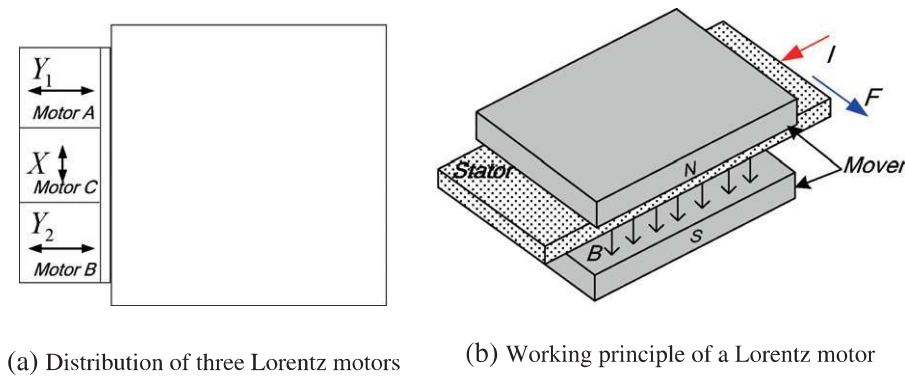


Fig. 3. The structural schematic diagram of the SSM.

of millimeters. The stator of the *LSM* is mounted on vibration-isolating supporting table, which is elaborately designed to isolate external vibration sources. The motion direction of the *LSM* is along the *Y* axis. The mover of the *LSM* is floated by four air-bearings in the *Z* direction and supported by two air-bearings on each side in the *X* direction, as illustrated in Fig. 2. Based on this configuration, the friction force becomes nearly zero. These air-bearings consist of pressure and vacuum zones to deliver high bi-directional stiffness. The electromagnetic force is produced by the interaction between the permanent magnetic field in the stator and the current in the mover, and the magnitude of the force depends on the strength of the permanent magnetic field.

The *SSM* is composed of three independent Lorentz motors, i.e. *X*, *Y1* and *Y2*, in an H-type of configuration, as shown in Fig. 3a, where the PM assembly is attached to the mover of the *SSM*, and the coil assembly to the stator. The stator and mover of the *SSM* are coupled by electromagnetic forces without mechanical contact. The stator of the *SSM* is connected with the mover of the *LSM* through a rigid driving arm, while the mover of *SSM* is floated by two annular air-bearings with high bi-directional stiffness. The positioning platen is levitated by rectangular air-bearings on a granite plate, and is connected on the mover of the *SSM* rigidly,

so that the *LSM* and *SSM*, combined together, drive the positioning platen in the *XY*-plane.

By combining these three Lorentz motors, the *SSM* has 3-DOFs, i.e. translation in the *X* and *Y* directions, and rotation about the *Z* axis. The maximum rotation of  $\pm 1.5$  mrad about the *Z* axis and translation of  $\pm 1.5$  mm in the *X* and *Y* directions for the positioning platen can be achieved by applying different currents in the *X*, *Y1* and *Y2* coils. Rotation is detected using a laser interferometer and corner mirrors which are mounted on the positioning platen.

Fig. 3b illustrates the working mechanism of a Lorentz motor. The magnetic field is formed by two PMs positioned opposite each other. The double-sided magnet array can be considered as a superposition of two symmetrical magnet arrays. Hence, magnetic flux lines are normal to the symmetry plane, which can be regarded as the surface of a yoke with high permeability. When a coil is placed in the magnetic field and a current is applied to the coil, Lorentz force is generated on the current-carrying wire loop. In this case, the coil assembly is the stator and the PMs assembly is the mover. Lorentz forces act perpendicular to the magnetic flux lines of an externally applied field. The force by which the PM assembly is pushed away is linear with the current through the coils of the stator.

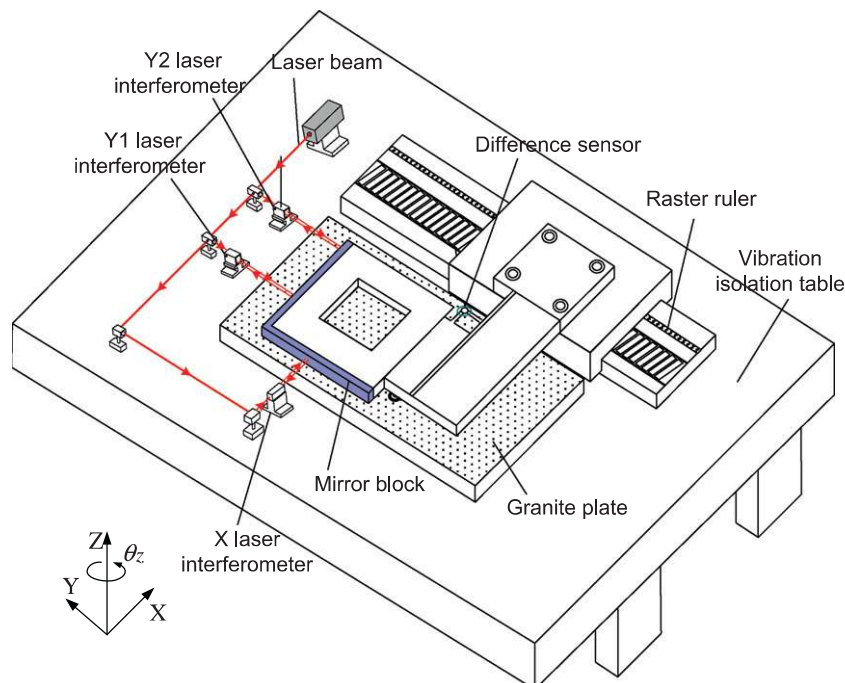


Fig. 4. Schematic view of the measurement system of the LPMS.

The three coils carrying a steady state current may cause severe heating to limit the force density, so the stator needs to be cooled.

## 2.2. Working mechanism of the LPMS

In order to localize the stage exactly, the LPMS is equipped with three kinds of positioning sensors with different resolutions, i.e. raster ruler, laser interferometer and Hall displacement sensor. The LSM can achieve a maximum displacement of 300 mm at a maximum speed of 1000 mm/s. The displacement of the mover of the LSM is measured by a raster ruler with a resolution of 1  $\mu\text{m}$ . The position and posture of the mover of the SSM is measured by the laser interferometer system, consisting of X, Y1 and Y2 laser interferometers. These laser interferometers and related optic elements are mounted along corresponding directions on the active vibration-isolating table, which provides a stable reference surface for the measurement system. Therefore, high positioning accuracy of the positioning platen can be achieved by the combination of these three laser interferometers. In our application, the measurement resolution of the laser interferometer is 0.6 nm. Three Hall displacement sensors, Y1, Y2 and X, are mounted on the stator of the SSM assembly to measure the relative movement of the positioning stage in the X, Y and Z directions, and further evaluate the motion synchronization offset between the SSM and the LSM. The Hall displacement sensor has the merits of low inertia and high response speed. The measurement resolution of the applied Hall displacement sensor is 1  $\mu\text{m}$ , and the measurement range is  $\pm 1.5$  mm with respect to the relative synchronization reference point of the LSM and the SSM. Fig. 4 depicts the schematic view of the measurement systems.

A master–slave control strategy is applied for LPMS motion control. The SSM acts as the master, and the LSM follows the SSM as the slave. As soon as the SSM starts moving, the Hall displacement sensors detect relative movement offset between the mover of the

SSM and the mover of the LSM, and send relative movement difference to the LSM controller. The LSM controller successively controls the LSM to follow the SSM. As a result, the mover of SSM will stay close to the center of the travel range of the SSM. The nanopositioning accuracy of the LPMS is further guaranteed by three laser interferometers in the SSM's motion stroke.

With such a configuration of positioning sensors, and using the master–slave control strategy, the LPMS can theoretically achieve a stroke of hundreds of millimeters with nanometer positioning accuracy. However, in order to bring such accuracy into reality, dynamic modeling of the LPMS is essential for controller design.

## 3. Dynamic modeling

### 3.1. Mass and inertia tensor

The total mass,  $M_1$ , of the assembled planar motion stage of the prototype, is 18 Kg, which includes movers of the SSM, the positioning platen, laser distance sensors, air distributor, and connection terminals. Given the whole geometric and assembly models of the stage, plus different material properties assigned to each part of the assembled platen, the center of mass (COM) of the stage can be obtained through a CAD tool, e.g. Pro/E. In this study, the moment of inertia matrix of the prototype in the unit of  $\text{Kg m}^2$  is

$$I = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{bmatrix} = \begin{bmatrix} 10.9 & -0.0047 & -0.0015 \\ -0.0047 & 0.9 & 2.2 \\ -0.0015 & 2.2 & 10.4 \end{bmatrix} \quad (1)$$

### 3.2. Equations of motion

Fig. 5 shows the simplified dynamic model of the LPMS. The LPMS can be separated into two rigid bodies:

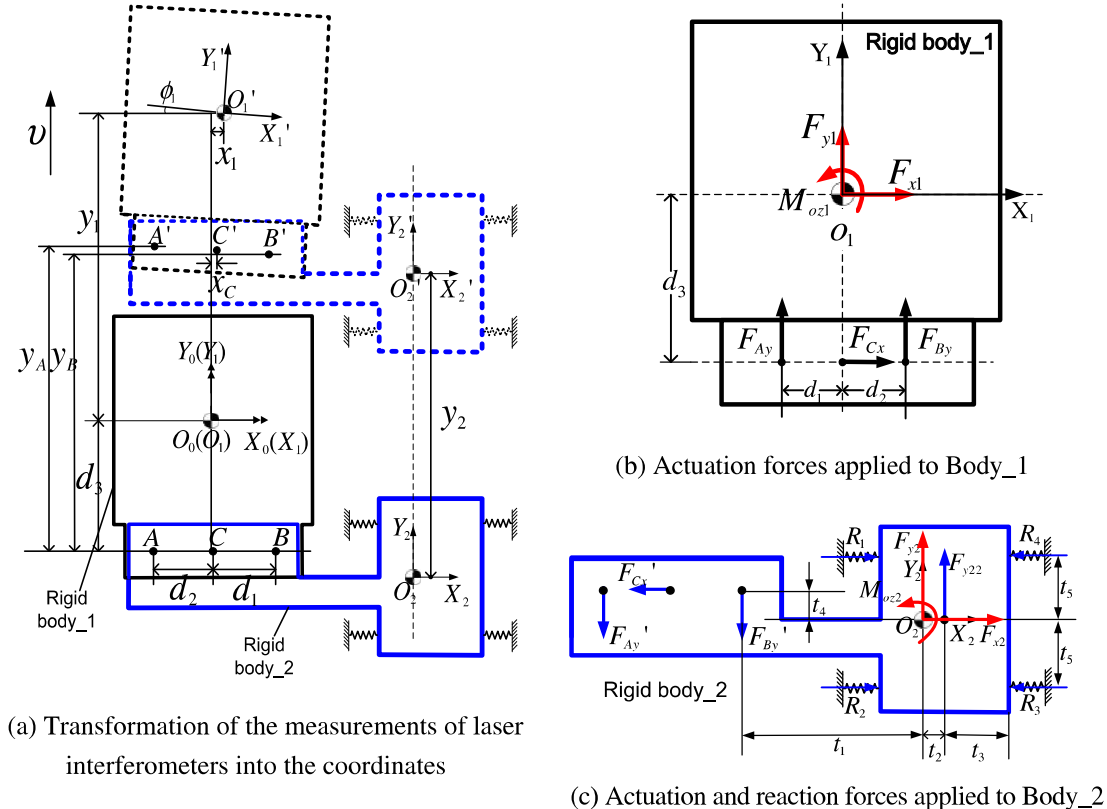


Fig. 5. Simplified dynamic model of the LPMS.



- (a) Body\_1 includes the mover of the SSM and positioning platen,
- (b) Body\_2 includes the stator of SSM, the driving arm and the mover of the LSM.

Two reference frames,  $X_1O_1Y_1$  and  $X_2O_2Y_2$ , are attached on the COMs of these two rigid bodies, respectively, and a global reference frame,  $X_0O_0Y_0$ , is defined such that it coincides with the frame  $X_1O_1Y_1$  at the initial state.

Because the action points of three Lorentz motors of the SSM do not coincide with the COM of the positioning platen as seen in Fig. 5b, the output Lorentz forces  $[F_{Ay} F_{By} F_{Cx}]$  of the SSM and the force/torque  $[F_{x1} F_{y1} M_{oz1}]$  at the COM of the planar positioning platen are related by the following equations

$$\begin{cases} F_{x1} = F_{Cx}(i_C) \\ F_{y1} = F_{Ay}(i_A) + F_{By}(i_B) \\ M_{oz1} = -F_{Ay}(i_A) \cdot d_1 + F_{By}(i_B) \cdot d_2 + F_{Cx}(i_C) \cdot d_3 \end{cases} \quad (2)$$

where  $d_1$  and  $d_2$  are the distances from the COMs of coils A and B, respectively, to the  $Y_1O_1Z_1$  plane, and  $d_3$  is the distance from the COM of coil C to the COM of stage.

Rewriting Eq. (2) into the matrix form, we obtain

$$\begin{pmatrix} F_{x1} \\ F_{y1} \\ M_{oz1} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ -d_1 & d_2 & d_3 \end{pmatrix} \begin{pmatrix} F_{Ay} \\ F_{By} \\ F_{Cx} \end{pmatrix} = \mathbf{P} \begin{pmatrix} F_{Ay} \\ F_{By} \\ F_{Cx} \end{pmatrix} \quad (3)$$

where

$$\mathbf{P} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ -d_1 & d_2 & d_3 \end{pmatrix}.$$

Then from Eq. (3), we obtain

$$\begin{pmatrix} F_{Ay} \\ F_{By} \\ F_{Cx} \end{pmatrix} = \mathbf{P}^{-1} \begin{pmatrix} F_{x1} \\ F_{y1} \\ M_{oz1} \end{pmatrix} = \begin{pmatrix} \frac{d_3}{d_1+d_2} & \frac{d_2}{d_1+d_2} & -\frac{1}{d_1+d_2} \\ -\frac{d_3}{d_1+d_2} & \frac{d_1}{d_1+d_2} & \frac{1}{d_1+d_2} \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} F_{x1} \\ F_{y1} \\ M_{oz1} \end{pmatrix} \quad (4)$$

The Lorentz force of the Lorentz motor is generated perpendicular to the magnetic field and the direction of the current if the coil of the motor is applied with a certain current, and the magnitude of the force follows the Lorentz's Law

$$F_{\text{lorentz}} = n \cdot B \cdot l \cdot i = k_f \cdot i \quad (5)$$

where  $B$  is magnitude of the magnetic field in the air gap of the motor in the unit of Tesla,  $i$  is the applied current in the coil of the motor in the unit of Ampere (A),  $k_f$  is the motor force constant,  $l$  is the effective length of the wire in the unit of meter, and  $n$  is the winding number of the coil of the motor.

For in-plane motions, a simplified system of Newtonian equations of motion (EOM) for the position of the platen,  $\mathbf{Q}_1 = [x_1, y_1, \phi_1]^T$ , can be expressed in the matrix form as

$$\mathbf{M}_1 \ddot{\mathbf{Q}}_1 = \mathbf{PK}_f \mathbf{I} \quad (6)$$

where  $\mathbf{M}_1 = \text{diag}[M_1, M_1, I_{zz}]$  is the mass and inertia matrix,  $\mathbf{K}_f = \text{diag}[k_{fA}, k_{fB}, k_{fC}]$  is the propulsion coefficient matrix, and  $\mathbf{I} = [i_A, i_B, i_C]^T$  is the coil current vector.

$\mathbf{Q}_1$  can be represented by the positions of three Lorentz motors  $\mathbf{Q}_1^{\text{prime}} = [x_C, y_A, y_B]^T$  as

$$\begin{pmatrix} x_1 \\ y_1 \\ \phi_1 \end{pmatrix} = \begin{pmatrix} 1 & \frac{d_3}{d_1+d_2} & -\frac{d_3}{d_1+d_2} \\ 0 & \frac{d_2}{d_1+d_2} & \frac{d_1}{d_1+d_2} \\ 0 & \frac{1}{d_1+d_2} & -\frac{d_1}{d_1+d_2} \end{pmatrix} \begin{pmatrix} x_C \\ y_A \\ y_B \end{pmatrix} \quad (7)$$

$\mathbf{P}$  gives rise to interactions among the propulsions in different driving directions. According to Kirchhoff's Law, the model of the electric circuit is represented as

$$\mathbf{U} - \mathbf{PK}_{\text{Bemf}} \frac{d\mathbf{X}}{dt} - \mathbf{L} \frac{d\mathbf{i}}{dt} - \mathbf{R}\mathbf{i} = 0 \quad (8)$$

where  $\mathbf{L} = \text{diag}[L_A, L_B, L_C]$  and  $\mathbf{R} = \text{diag}[R_{Ay}, R_{By}, R_{Cx}]$  are the inductance and resistance matrices of the coil, respectively,  $\mathbf{U} = [u_{Ay}, u_{By}, u_{Cx}]^T$  is the input voltage vector for the motor, and  $\mathbf{K}_{\text{Bemf}} = \text{diag}[K_{\text{Bemf}_A}, K_{\text{Bemf}_B}, K_{\text{Bemf}_C}]^T$  is the backward electro-motive force constant matrix. From Eq. (8), we can see that the applied current is related to several complex factors.

Similarly, the action point of the mover of the LSM does not coincide with the COM of the rigid body\_2 as seen in Fig. 5c, and the driving force  $F_{Dy}$  of the LSM, counterforce  $[F'_{Ay} F'_{By} F'_{Cx}]$  and the force/torque  $[F_{x2} F_{y2} M_{oz2}]$  at the COM of the rigid body\_2 are related by the following equations

$$\begin{cases} F_{x2} = R_1 + R_2 - R_3 - R_4 - F'_{Cx} \\ F_{y2} = F_{Dy} - F'_{Ay} - F'_{By} \\ M_{oz2} = F'_{Ay} \cdot (d_1 + d_2 + t_1) + F'_{By} \cdot t_1 + F'_{Cx} \cdot d_4 + F_{Dy} \cdot t_2 \\ \quad + (R_2 + R_4 - R_1 - R_3) \cdot t_5 \end{cases} \quad (9)$$

where  $R_i (i = 1, 2, 3, 4)$  denotes carrying capacities of the four air-bearings in the  $X$  direction,  $t_i (i = 1, 2, \dots, 5)$  is defined as the distance between the COM of rigid body\_2 and forces (in Fig. 5c) for the four air-bearings.

The LSM has only one DOF, measured by the raster ruler in the  $Y$  direction, and stiffness of the air-bearings in  $X$  direction is very high (50–100 N/ $\mu\text{m}$ ). Therefore, the motions in the  $X$  and  $R_z$  directions can be ignored for Body\_2. A simplified dynamic equation for the position  $\mathbf{Q}_2 = [y_2]$  of the Body\_2 can be expressed as

$$\mathbf{M}_2 \ddot{\mathbf{Q}}_2 = k_{fD} i_D - k_{fA} i_A - k_{fB} i_B \quad (10)$$

where  $i_D$  is the applied current in the coil of the LSM in the unit of Ampere (A), and  $k_{fD}$  is a force constant of the LSM.

The LPMS is propelled by linear Lorentz forces through three Lorentz motors. These Lorentz forces are considered to be linear with electric currents in our model. It is essential to analyze the relations among Lorentz forces, magnetic fields, currents and displacements. However, it has been well-recognized to be difficult to experimentally obtain detailed information about electromagnetic forces as well as magnetic fields due to small sizes of the air gap and the lack of proper measuring device, therefore numerical investigation is carried out.

#### 4. Analysis of electromagnetic field and Lorentz force with 3-D FEM

##### 4.1. 3-D FEM analysis model

To achieve nanometer positioning accuracy, higher precision modeling is required for controller design. In the LPMS, end effects, magnetic field coupling and material nonlinearity are amounts which influence cannot be neglected comparing to the expected accuracy. In this paper, 3-D finite element method (3-D FEM) is applied to analyze the magnetic field. The advantage of using 3-D FEM is the ease when dealing with complex geometries and magnetization direction of the permanent magnet. In addition, end effects and nonlinearity of materials can be considered in the FE computation of magnetic fields.

The SSM consists of a compact arrangement of three Lorentz motors in a plane, to generate high force densities with small leakage magnetic fluxes. The 3-D FE model is achieved as shown in Fig. 6. Simulation parameters are listed in Table 1.

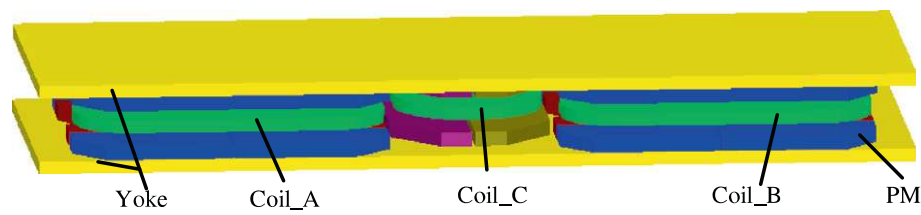


Fig. 6. 3-D FE electromagnetic model of the SSM.

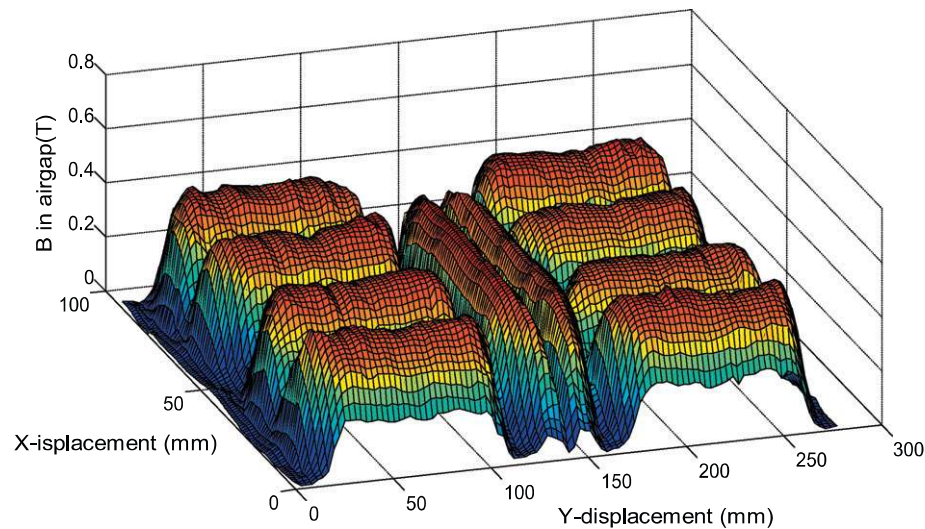
**Table 1**  
Simulation parameters of the SSM.

| Parameter                           | Units | Value                      |
|-------------------------------------|-------|----------------------------|
| Coils A and B                       | mm    | $98 \times 86 \times 4$    |
| PM for coils A and B                | mm    | $100 \times 90 \times 4.5$ |
| Coil C                              | mm    | $71 \times 47 \times 4$    |
| PM for coil C                       | mm    | $75 \times 51 \times 4.5$  |
| Yoke                                | mm    | $280 \times 90 \times 4$   |
| Air-gap                             | mm    | 1.5                        |
| Coercive force of PM                | KA/m  | 821                        |
| Remnant magnetic flux density of PM | T     | 1.35                       |

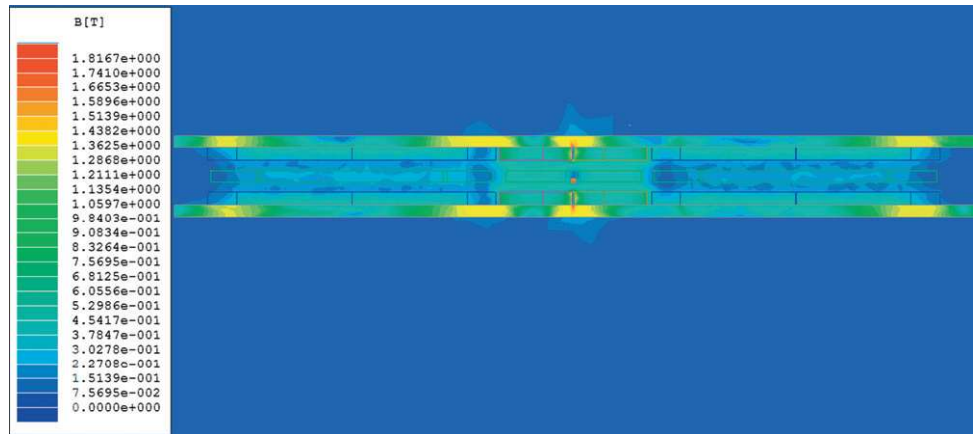
Under the condition of high permeability and small air gaps, the magnetic field of the SSM is analyzed based on the electromagnetism theory. The adaptive analysis method is adopted to compute the magnetic field, and more accurate results is achieved by breaking down the smaller tetrahedron in the mesh. The truncation error in the numerical computation is set to be 1%.

4.2. Calculation of the magnetic field in the air gap

Fig. 7 shows the distribution of magnetic flux density in XY-plane and XZ-plane of the SSM, when no current is applied on the



(a) The middle of the air-gap in XY-plane



(b) XZ-plane of the SSM

Fig. 7. Cross-sectional view of the magnetic flux density in the SSM.

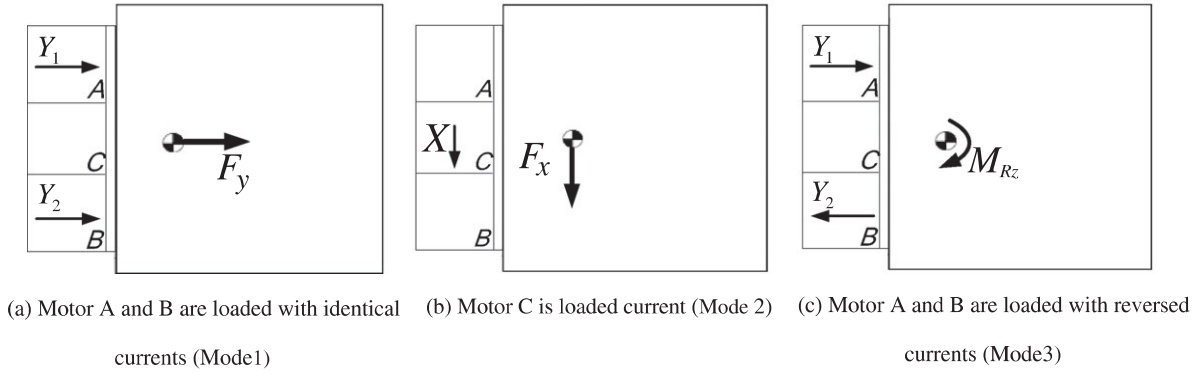


Fig. 8. Three loading modes of the SSM.

coils A, B and C. It is clear from Fig. 7a that, the maximum magnetic flux density is 0.53 T in the middle of the air gap, while that of the mover's yoke is rather large in the range of 1.2–1.8 T in Fig. 7b.

#### 4.3. Analysis of the Lorentz force–current relation

The Lorentz force–current relation is the main characteristic of the SSM. To derive the Lorentz force of the Lorentz motor from the FE computation, the Maxwell stress method is employed. In this method, the stress must be integrated over the overall surface of the mover to obtain the actuating force.

The Lorentz force acting on a current-applied coil can be calculated given the distribution of the flux density  $\mathbf{B}$  on a closed envelope surface  $S$  around that coil [25]

$$\mathbf{F} = \frac{1}{\mu} \oint_S \mathbf{T} d\mathbf{s} \quad (11)$$

where  $\mathbf{T}$  is the Maxwell stress tensor

$$\mathbf{T} = \begin{pmatrix} B_x^2 - \frac{1}{2}|\mathbf{B}|^2 & B_x \cdot B_y & B_x \cdot B_z \\ B_y \cdot B_x & B_y^2 - \frac{1}{2}|\mathbf{B}|^2 & B_y \cdot B_z \\ B_z \cdot B_x & B_z \cdot B_y & B_z^2 - \frac{1}{2}|\mathbf{B}|^2 \end{pmatrix} \quad (12)$$

The Maxwell stress method is quite sensitive to the choice of the sectional plane in the air gap.

The SSM has three loading modes, as shown in Fig. 8a–c. In **Mode 1** as shown in Fig. 8a, identical currents are applied on Motor A and B, and translation in the Y direction is achieved; In **Mode 2** as shown in Fig. 8b, current is applied only on motor C; In **Mode 3** as

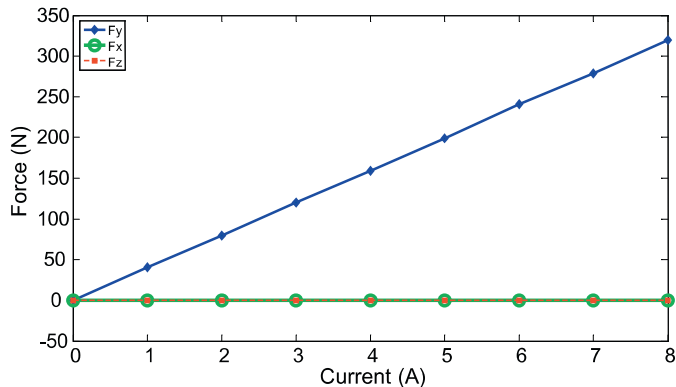


Fig. 9. Magnetic forces as a function of the current, in the X, Y and Z directions in Mode 1.

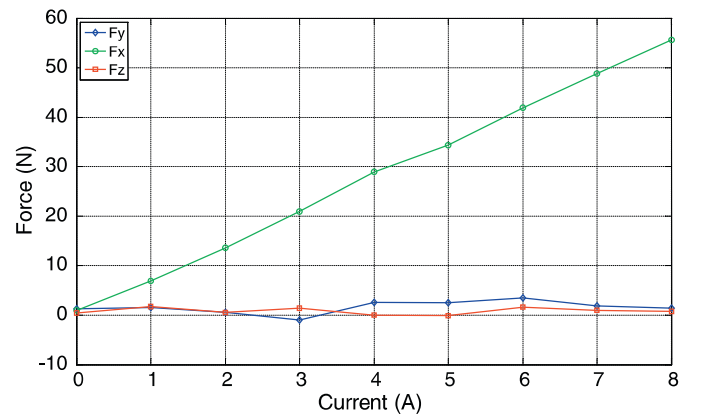


Fig. 10. Magnetic forces as a function of the current, in the X, Y and Z directions in Mode 2.

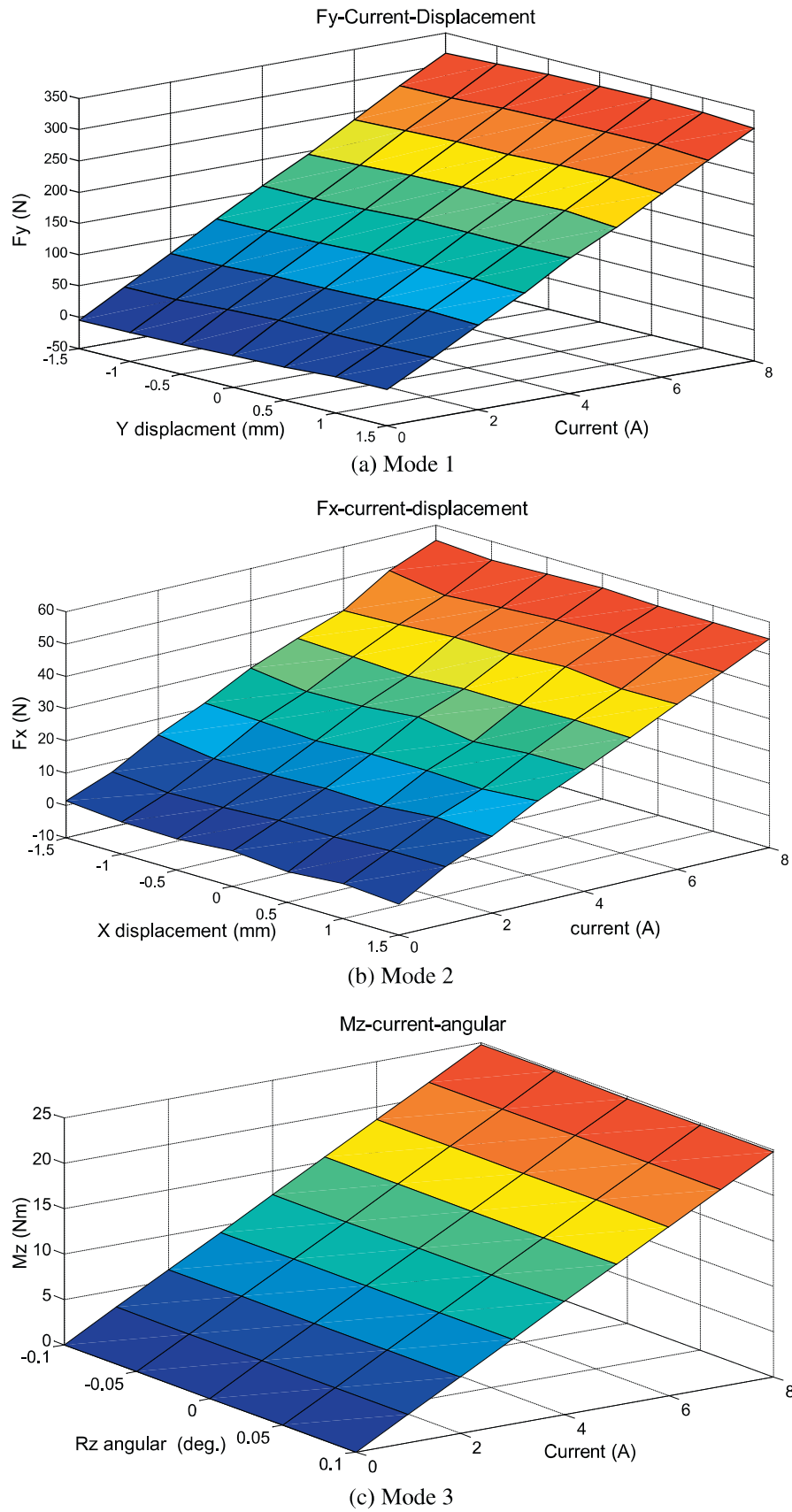
shown in Fig. 8c, when reversed currents are applied on coils A and B, the stage rotates about the Z axis.

Figs. 9 and 10 indicate the relations between forces in X, Y and Z directions and currents in coils. From the results of 3-D FE analysis shown in Figs. 9 and 10, it is clear that propulsion increases almost linearly with the increase of the current. Magnetic forces experience slight fluctuation in the Y and Z directions in Fig. 10. This is mainly caused by the coupling of magnetic fields induced by the three Lorentz motors under different applied currents. The three Lorentz motors are not far enough from each other in distance so that their magnetic fields interact with each other. Therefore, the magnetic field coupling of three Lorentz motors should be considered in controller design for the stage.

#### 4.4. Relation between the Lorentz force and displacement

The Lorentz motors are excited by given DC currents so that the positioning stage can generate motions in X, Y and  $R_z$  directions. The displacement range of the SSM between the stator and mover is  $\pm 1.5$  mm in the X and Y directions, and the maximum rotation is  $\pm 1.5$  mrad about the Z axis.

Figs. 11a–c plot the magnetic forces as a function of the current and displacements in the Y, X, and  $R_z$  directions, respectively. Thrust ( $F_x$  and  $F_y$ ) and torque  $M_z$  increase linearly with increasing currents and show small fluctuations with displacements in the three motion directions as shown in Fig. 11a–c. From the comparative analysis of the three figures, it is illustrated that the fluctuations of magnetic forces are closely related to translation in the X direction. If these coils are positioned in the middle of the air



**Fig. 11.** Relations among electromagnetic forces, currents and displacements in each mode.

gap, the bent magnetic lines are also perpendicular to the coils, which results in horizontal component forces. However, if the coils

deviate from the position in the middle of the air gap, a vertical component is produced as shown in Fig. 12.



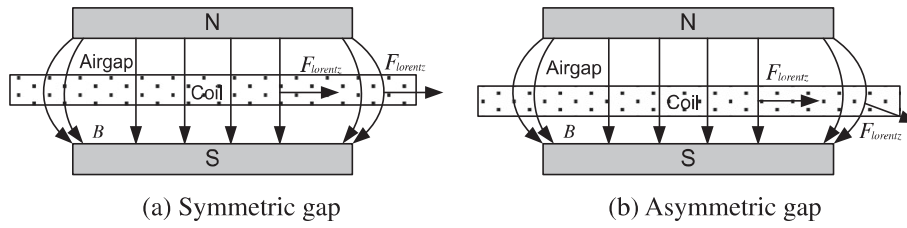


Fig. 12. Magnetic flux lines and Lorentz forces.

Therefore, in order to obtain steady thrust forces, three coils should be laid in the middle of air gap between the PMs, to avoid disturbance in the  $Z$  direction. Moreover, Lorentz forces fluctuation results from non-uniformity of the magnetic field, coupling of magnetic fields and large displacements in the  $X$ ,  $Y$ , and  $R_z$  directions, which should be compensated in controllers as static errors.

### 5. Mechanical simulation and analysis of the LPMS

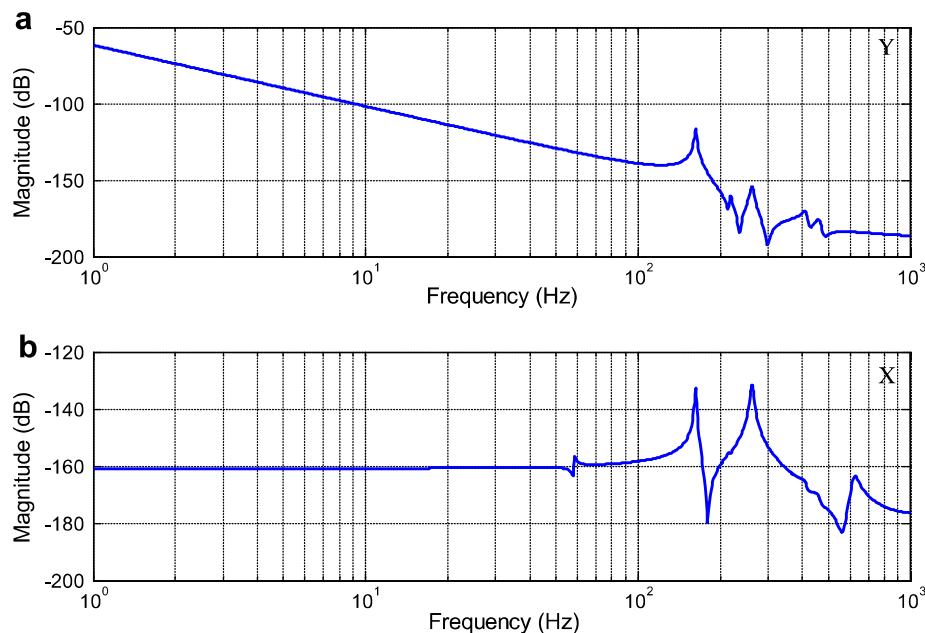
In order to determine dynamic performances in the  $X$ ,  $Y$  and  $R_z$  directions, frequency responses of the LSM and SSM in motion directions are investigated. According to the built dynamic model, we use an equivalent force input which is normalized in amplitude, since we are only interested in the relative accelerations at various frequencies. The input signals are force excitations and the output signals are displacements on the COM. Fig. 13 plots the simulated frequency response functions (FRFs) of the LSM when a swept sinusoidal excitation in the  $X$  and  $Y$  directions is applied on the mover of the LSM. As shown in Fig. 13, the system shows a double integral behavior in a large low-frequency range and some additional high-frequency resonances. It is also shown that three peak value points correspond to frequencies of 162 Hz, 262 Hz, and 462 Hz. Due to these peak value points near the bandwidth of the system, controllers should be designed to avoid these effects.

Fig. 14 shows the FRFs of the SSM in the  $X$ ,  $Y$  and  $R_z$  directions. Fig. 14a demonstrates that two peak value points correspond to frequencies of 309 Hz and 526 Hz, and Fig. 14b demonstrates

one peak value point corresponding to the frequency of 284 Hz. Due to the high stiffness of air-bearings and small motion masses, these peak value frequencies are far from the bandwidth of the system.

### 6. Experiments of mechanical FRFs

To obtain frequency responses of the LPMS in three motion directions, we perform mechanical FRFs tests using the dynamic signal analyzer (DSA). The operational pressure of air-bearings is in the range of  $5.62 \times 10^5$ – $6.64 \times 10^5$  Pa. Fig. 15 presents mechanical FRFs of the SSM in the  $X$ ,  $Y$  and  $R_z$  directions. With force excitations as input signals and measured positions of the mover as output, the system demonstrate double integral behaviors in a large low-frequency range and high-frequency resonances as shown in Fig. 15. The mechanical FRFs in the low-frequency range are approximately straight lines in the magnitude plot with a slope of 40 dB/decade, and horizontal lines in the phase plot starting from the  $-180^\circ$ . In other words, these phases of low-frequency fields are  $-180^\circ$  in open loop systems. This is the result of the frictionless motion using air-bearings in the  $X$ ,  $Y$  and  $R_z$  directions, and Newton's Second Law is satisfied between the driving force and displacement. The phase delay  $180^\circ$  need to be compensated to ensure the stability of servo system. We can clearly see in Fig. 15 that there are unstable zeros about 750 Hz in phase diagrams along the  $X$ ,  $Y$  and  $R_z$  directions. Because the action point of driving forces of the SSM does not coincide with the COM of the positioning platen,

Fig. 13. Simulated FRFs of the stator of the SSM in  $Y$  (top sub-figure) and  $X$  (bottom sub-figure) directions, when excited on the mover of the LSM.

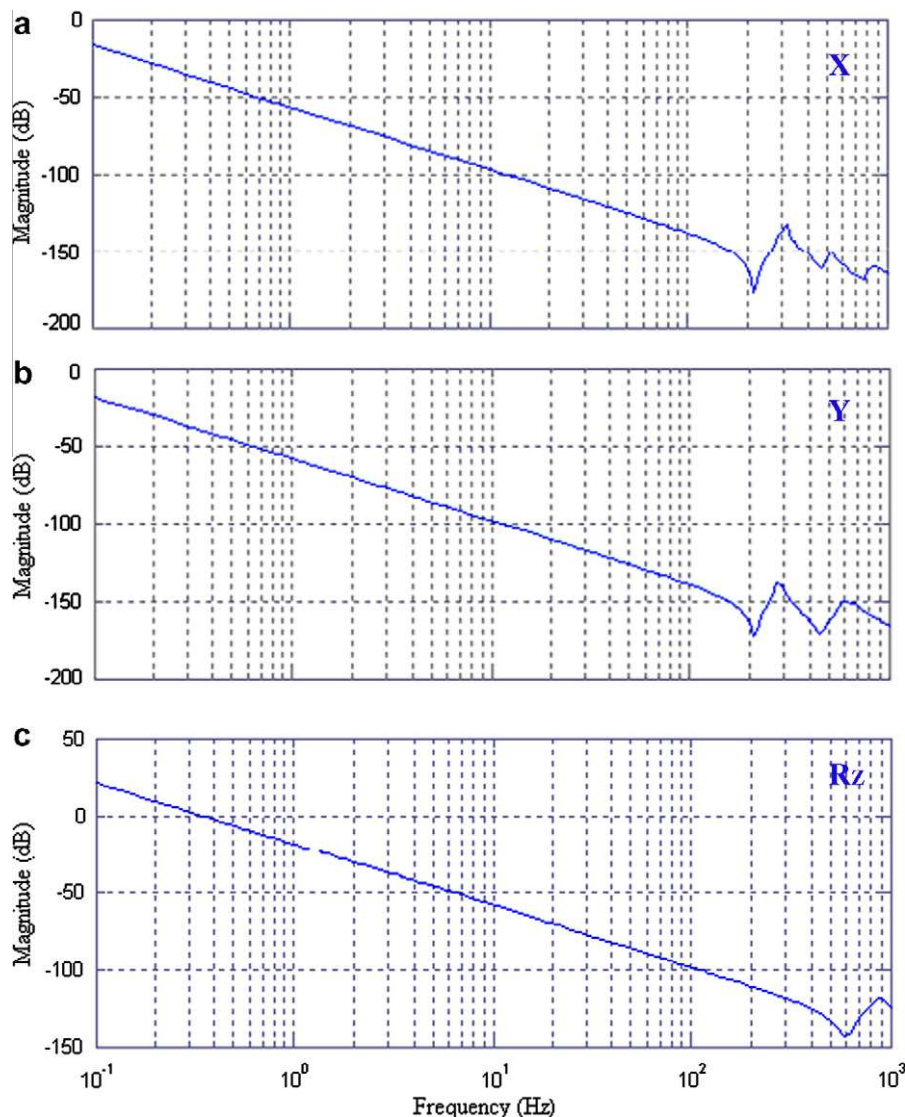


Fig. 14. FRFs of the SSM in the X (top sub-figure), Y (middle sub-figure) and  $R_z$  (bottom sub-figure) directions when excited on the COM of the SSM.

and connections between rigid bodies are flexible, these fast complex zeros can appear in the phase diagram. This phenomenon is called “actuator flexibility”, which can induce phase delay. Therefore, these zeros should be compensated in controllers. From Fig. 15, it is reasonable to conclude that when ignoring structural resonances of high frequency of the system, the stage can be modeled as a free moving mass. Therefore, it is effective to design the feed forward signal with only an acceleration profile. Comparing Figs. 14 and 15, we can see very good agreement between the simulation results and test data, which means that we can easily simulate and estimate total scanning accuracy based on the stage simulation results.

## 7. Positioning performance

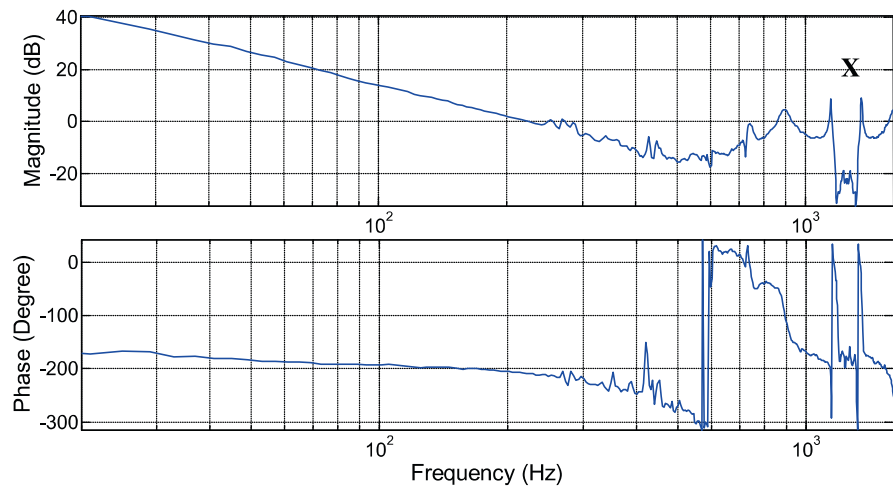
In our current implementation, a motion control strategy, combining basic PID control, master–slave control and feed forward control, is adopted. Fig. 16 shows a block diagram of the control strategy.

The controllers of the LSM and SSM are separately proposed, designed, and embodied in the LPMS. PID and low-pass filter control-

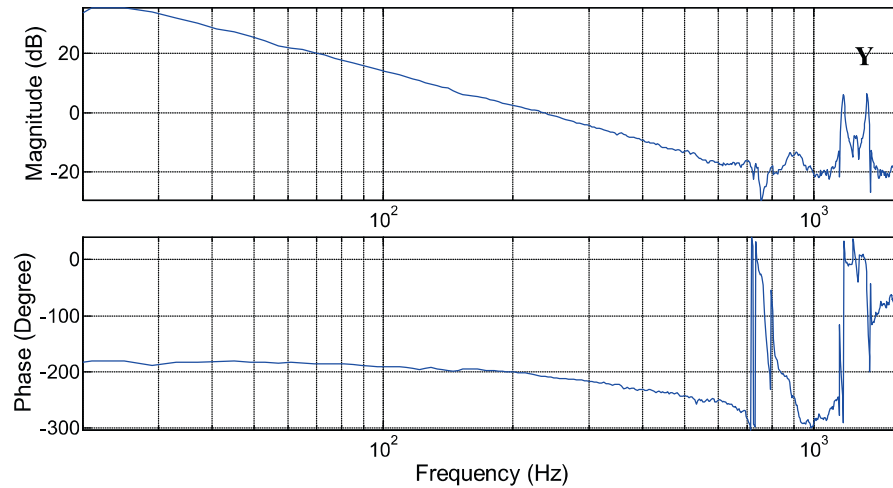
lers are applied as a basic tracking controller of the LSM and SSM, which is to stabilize the mechanism and create maximum robustness and phase margin. In our master–slave control, the SSM movement is the master and the LSM follows as the slave. According to the results of experimental modal analyses, the notch filters are adopted to achieve phase margin in the control loop in the neighborhood of a (mechanical) resonance frequency. Compensation of fluctuations in Lorentz forces due to position change and magnetic field coupling is considered in the design of controllers.

Tracking simulation results are shown in Fig. 17 along the X direction. As can be seen, the SSM moves in the region of  $2\ \mu\text{m}$ , which means the short stroke motor moves in a very limited range with high accuracy, while the LSM follows the short stroke over a long range with relatively low accuracy. As can be seen in Fig. 17, it takes about 130 ms to reach 10 nm. This desired motion control system can achieve both high speed and ultra precision.

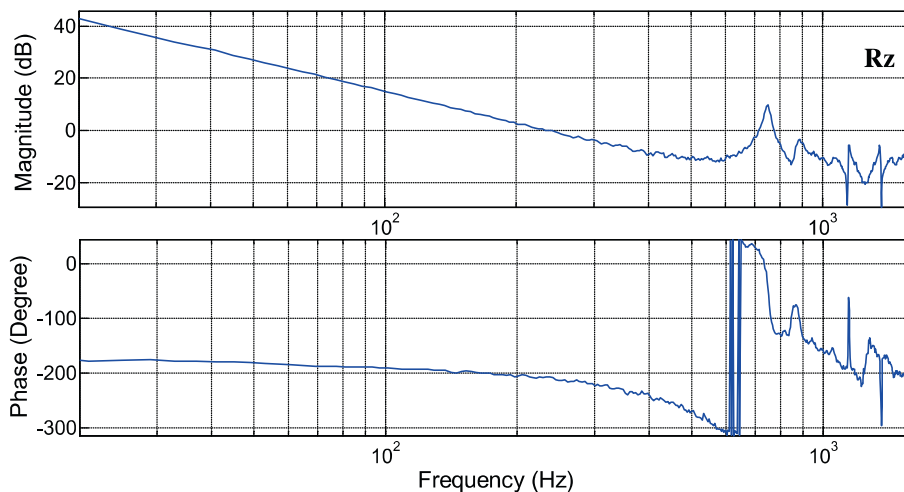
Experiments are conducted many times to obtain positioning accuracy (PA) of the positioning stage. According to the demand of three  $\sigma$  index, PA of the stage can be evaluated by moving average (MA) and moving standard deviation (MSD). These following three steps are adopted.



(a) X direction



(b) Y direction



(c) Rotation in Z axis

Fig. 15. Mechanical FRFs of the SSM by experiments.

**Step 1:** Obtain the positioning errors (*PEs*) in the *X*, *Y*, and *R<sub>z</sub>* directions;

**Step 2:** Compute moving average (*MA*) and Moving Standard Deviation (*MSD*);

**Step 3:** Compute positioning accuracy of the stage.

To obtain the *PEs* in the *X*, *Y*, and *R<sub>z</sub>* directions in **Step 1**, a number of experiments have been conducted, and a large amount of

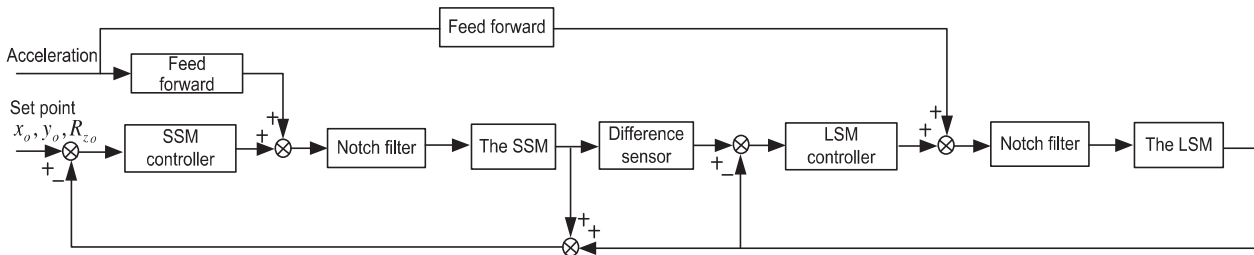


Fig. 16. Block diagram of the master-slave control.

data are acquired. The procedure of experiments for *PEs* is as follows:

- (1) The stage is positioned on the initial point  $O(0, 0)$ ;
- (2) The stage moves to the point  $O'(0, 50 \text{ mm})$  at a speed of  $120 \text{ mm/s}$ ;

- (3) The actual positions in the  $X$ ,  $Y$  and  $R_z$  directions at point  $O'$  are measured by the laser measuring system, and on-line recorded with laser counting-cards; and the measurement and data acquisition procedure repeat for 1000 times at point  $O'$ ;
- (4) These 1000 groups of sample points are exported from laser counting-cards to a host computer for off-line accuracy assessment.

**Step 2** is to compute the *MA* and *MSD* by using *PEs* in the  $X$ ,  $Y$ , and  $R_z$  directions. There are two important indices for computing *PA* of the stage. They are defined as

$$MA = \frac{1}{T} \int_0^T PE(t) dt \quad (13)$$

$$MSD = \sqrt{\frac{1}{T} \int_0^T (PE(t) - MA)^2 dt} \quad (14)$$

where  $T$  is the total time,  $PE(t)$  is the position error at time  $t$ . The *MA/MSD* is defined as the mean/ $1\sigma$  value of the positioning error during the period  $T$ .

**Step 3** is to compute the *PA* of the stage using the *MA* and *MSD*. The *PA* can be expressed as

$$PA = MA + 3 \times MSD \quad (15)$$

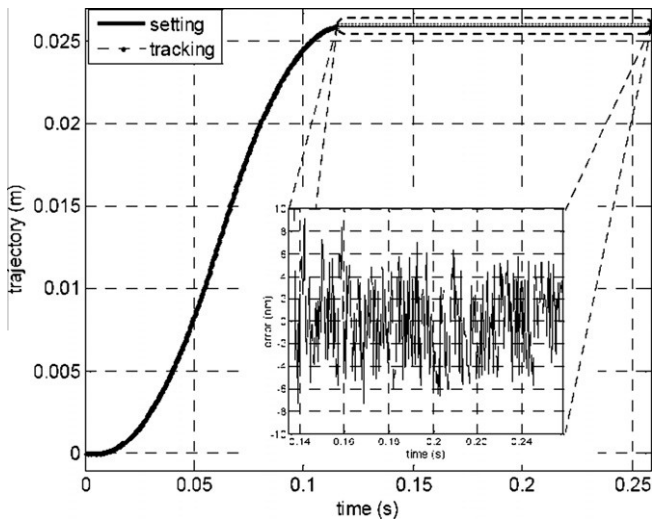
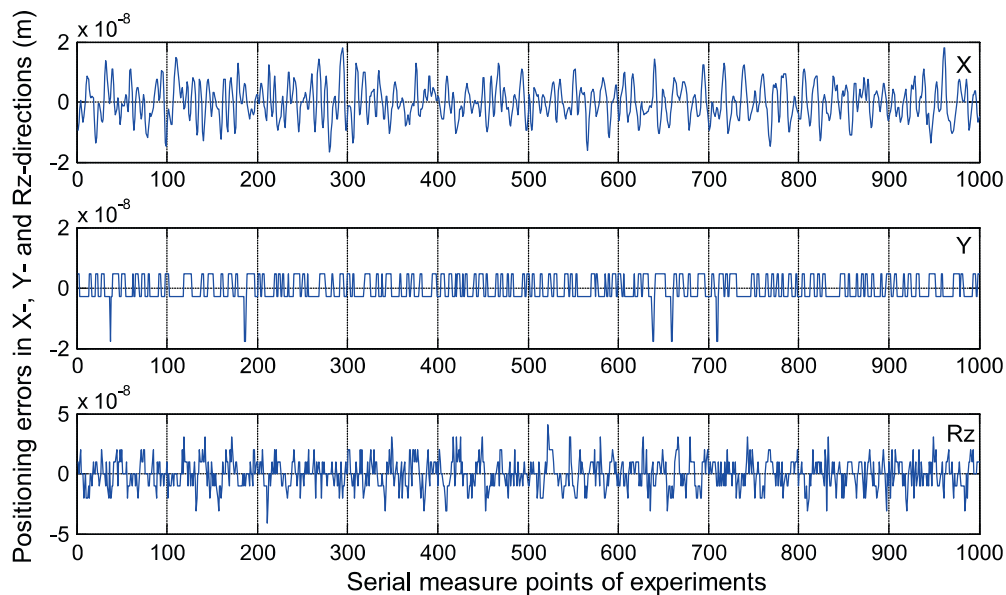


Fig. 17. Simulated tracking errors of the LPMS.

Fig. 18. Positioning errors in the  $X$  (top sub-figure),  $Y$  (middle sub-figure), and  $R_z$  (bottom sub-figure) directions.



**Table 2**

Positioning accuracy of the LPMS.

|         | X              | Y              | $R_z$          |
|---------|----------------|----------------|----------------|
| MA (m)  | +1.760862e–011 | +6.108950e–012 | +3.568062e–011 |
| MSD (m) | +4.931302e–009 | +3.943785e–009 | +1.227973e–008 |
| PA (m)  | +1.481151e–008 | +1.183747e–008 | +3.687486e–008 |

Fig. 18 shows the experimental results of  $PEs$  in the  $X$ ,  $Y$ , and  $R_z$  directions. The results of  $MA$ ,  $MSD$  and  $PA$  are listed in Table 2. From Fig. 18 and Table 2, the  $PA$  of the LPMS reaches 15 nm.

## 8. Conclusions

In this paper, a prototype of 3-DOF LPMS with high speed, long stroke, and nanometer positioning precision is modeled and analyzed. The following conclusions can be drawn.

- (1) An approach combined Newtonian formulas to establish dynamic model and numerical simulations to obtain electro-magnetic forces and mechanical FRFs is presented to analyze of a 3-DOF LPMS for nanopositioning.
- (2) Good linear relationship presents between driving forces and electric currents. Displacements in the  $X$ ,  $Y$  and  $R_z$  directions have effects on thrust forces in a limited working stroke, and the resulting fluctuations are compensated in controllers. The magnetic coupling among the three Lorentz motors are weak, based on which independent control of the three Lorentz motors can be easily realized.
- (3) According to dynamic simulation and mechanical FRFs experiments, the system shows a double integral behavior in a large low-frequency range and some additional high-frequency resonances in  $X$ ,  $Y$  and  $R_z$  directions. The FRFs of the LPMS obtained by numerical simulation are in good agreement with those obtained experimentally.
- (4) The control system combines basic loop control, master-slave control and feed forward control. Simulation and experimental results confirm 15 nm positioning accuracy of the LPMS.

## Acknowledgments

This work is financially supported by the Major State Basic Research Development Program of China (973 Program) under Grant Number of 2009CB724205, the Important National Science & Technology Specific Projects (2009ZX02304, 2009ZX02308), and the Natural Science Foundation of China (NSFC) under Grant Number of 50605025.

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